

Does Delhi Build Schools Where Children Live?

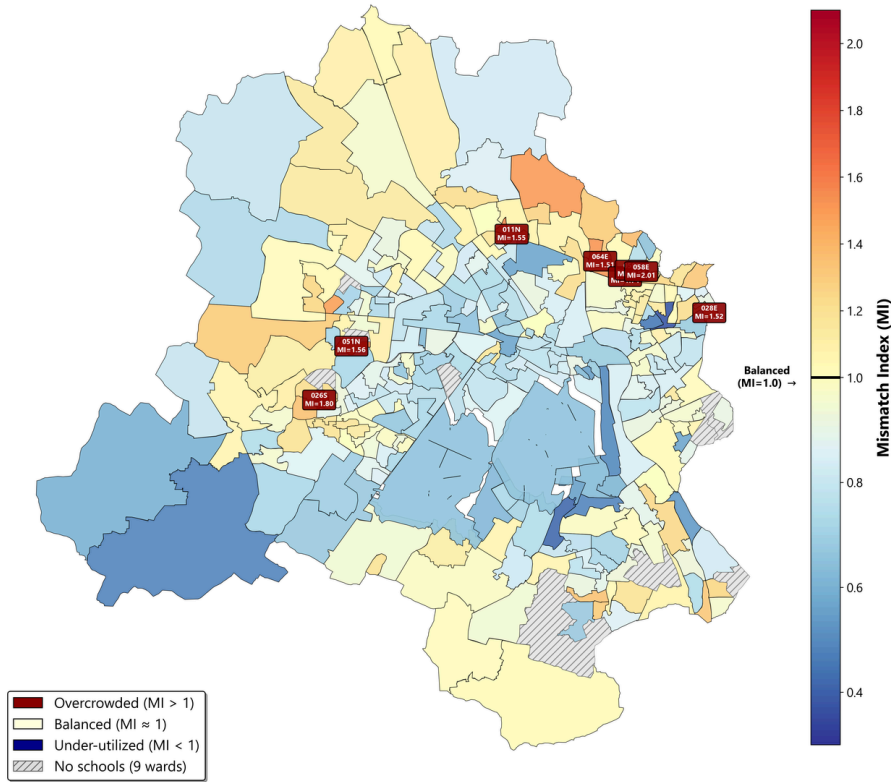
Spatial Mismatch Between Government School Capacity and Child Population Across Delhi's MCD Wards

The Spatial Illusion vs. Reality

- **The Illusion:** Delhi has ~1.47M government school seats for ~1.5M enrolled students. On a city-wide average, capacity appears sufficient.
- **The Reality:** Empty seats in wealthy South Delhi provide no relief to overcrowded children in North Delhi. Education is a spatial problem.
- **Objective:** Quantify the ward-level geographic mismatch, prove it is spatially clustered, and measure the spillover of the crisis using spatial econometrics.

The Baseline Mismatch

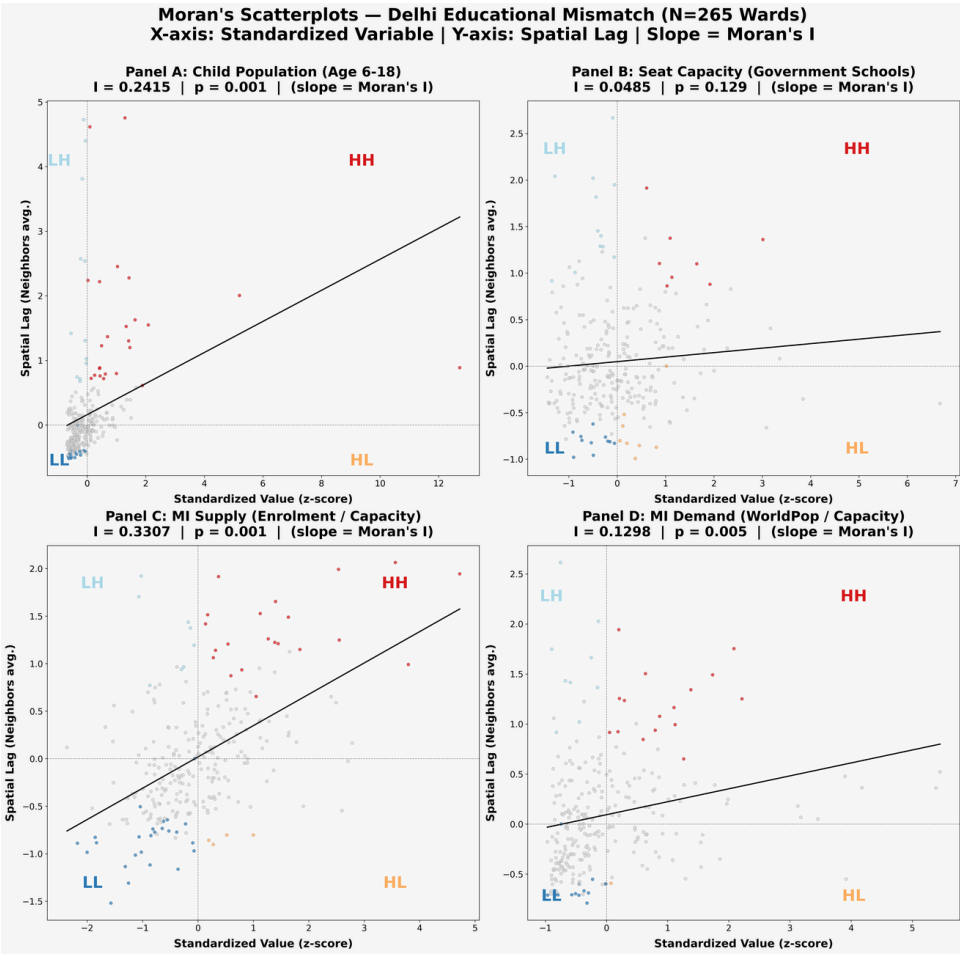
- **Mismatch Index (MI):** Total Enrolment / (Classrooms × 40)
- Rule: MI > 1.0 = Overcrowded. MI < 1.0 = Surplus



Methodology Flow

- **ESDA:** Moran's I, LISA, Variogram to prove spatial dependence.
- **Global Model:** OLS fails → LM Diagnostics select SAR to quantify spillover.
- **Local Model:** GWR to map geographic variations in crisis drivers.

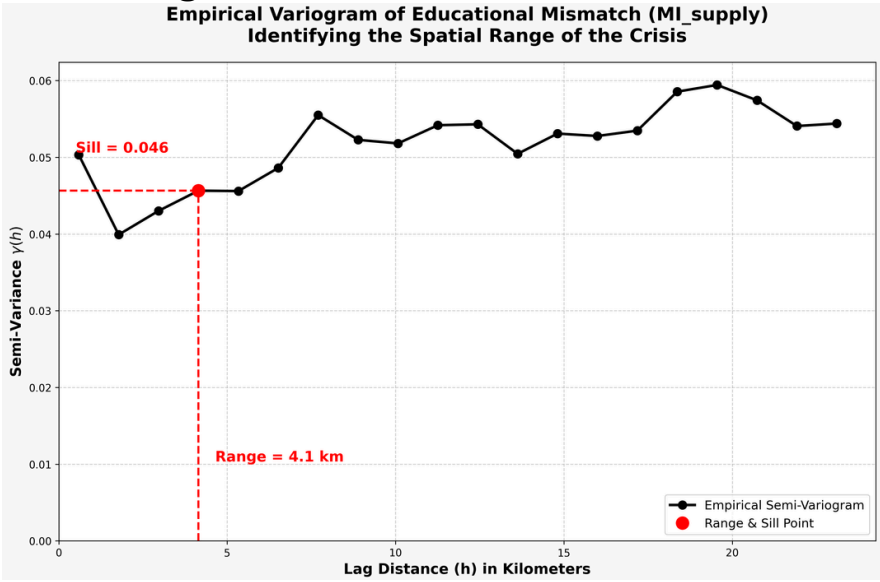
Proving Spatial Clustering



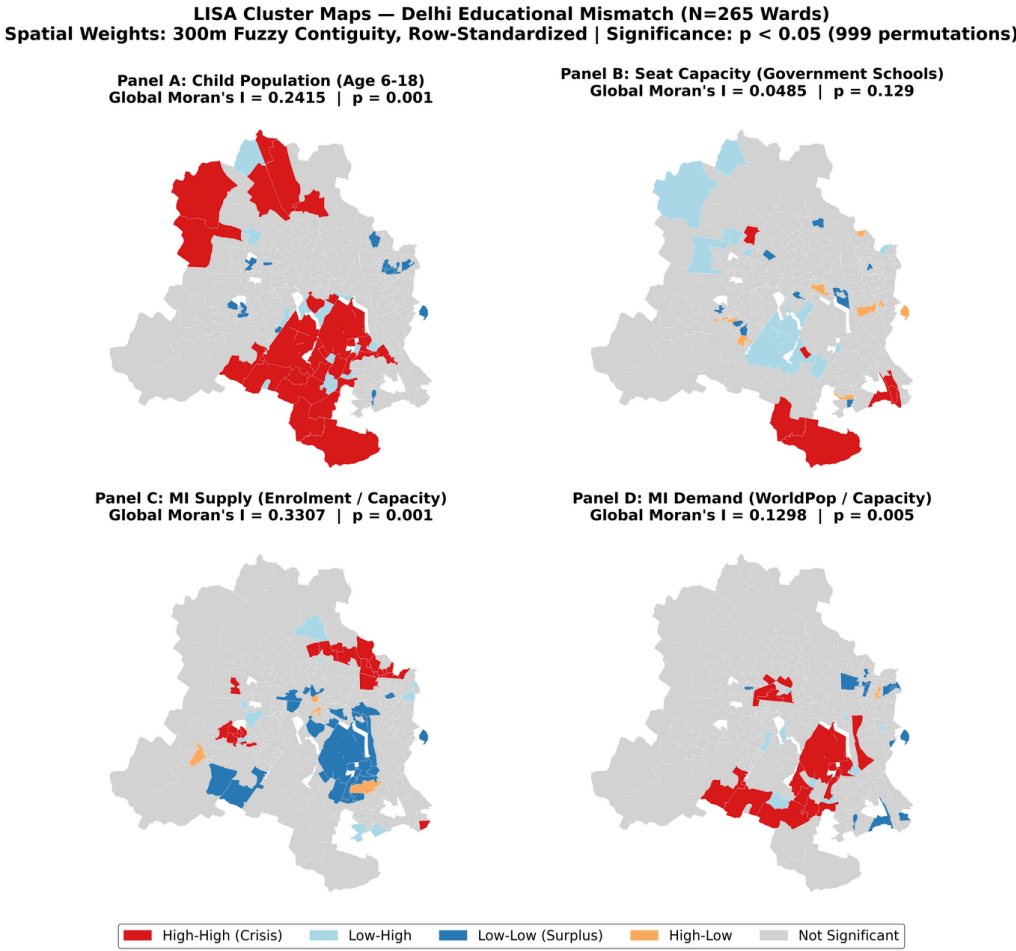
Moran's I confirms severe clustering across 4 metrics. **Note:** MI Supply measures actual school enrolment, while MI Demand uses WorldPop satellite estimates.

The 4.15km Barrier

The variogram proves the spatial crisis stretches for 4.15 km. A child must travel over 4km to escape the overcrowding.



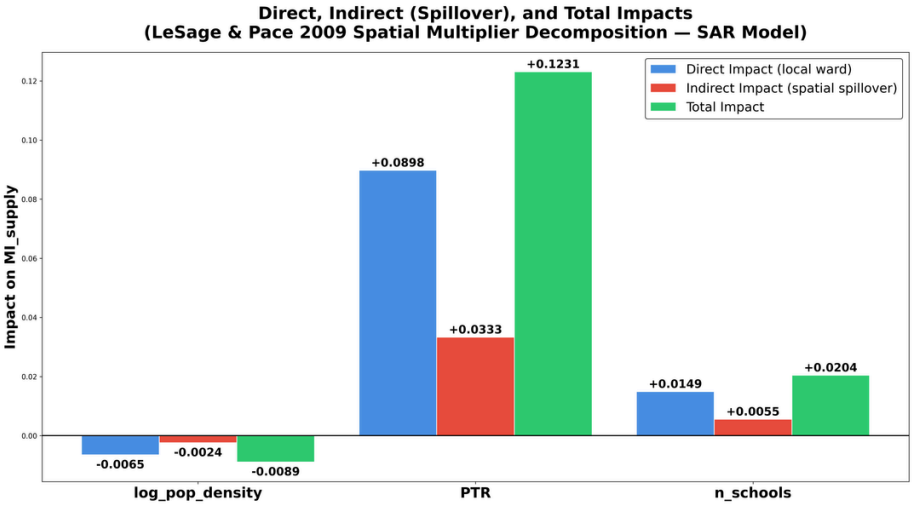
Pinpointing the Clusters



14 overcrowded wards in North/East (Red) and 23 surplus wards in South (Blue).

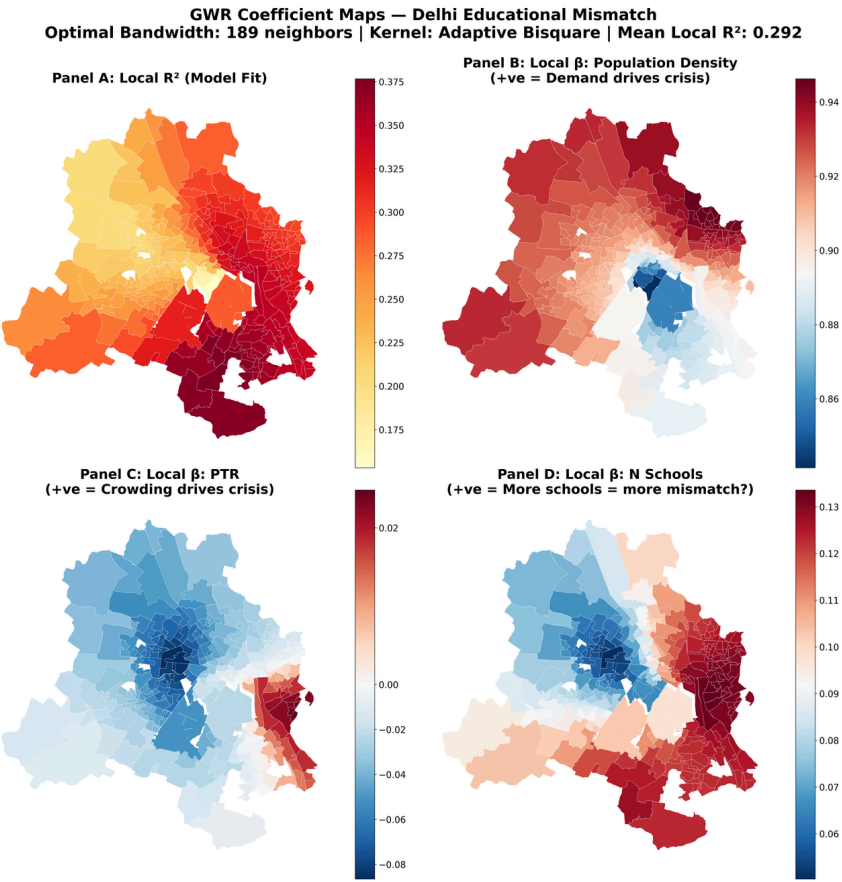
The Spatial Multiplier Effect

Because mismatch is spatially clustered, OLS fails → LM diagnostics select SAR, quantifying network spillovers (**p = 0.2855, p < 0.01**) and revealing a 39.8% spillover where local teacher shortages increase overcrowding across neighbouring wards.



Geographic Non-Stationarity

SAR is a city-wide average. GWR maps how the crisis drivers change locally.



- **The PTR Paradox:** GWR proves East Delhi mismatch is driven by severe teacher shortages. However, in North Delhi, PTR is insignificant.
- **The Invisible Population:** North Delhi is overcrowded due to massive, unmapped populations. WorldPop satellite data missed these people due to Vertical Density Blindness, Census Averaging/Spatial Smoothing, and Unrecognised Settlement Exclusion.

Policy Implications

- **Targeted Relief:** East Delhi needs rapid deployment of teachers. North Delhi needs physical land/school construction.
- **Spillover Benefit:** Adding teachers to one ward provides a 39.8% larger benefit by relieving cross-border overcrowding pressure.
- **Data Warning:** Urban planners cannot blindly rely on global satellite population data for Indian informal settlements.